

ABHISHEK SINGH

AI Engineer · Generative AI Lead · LLM Systems & RAG Architecture · Agentic Orchestration · LLMOps
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\$500K+	90%	99.9%	3.5 Yrs
Revenue Recovered	Manual Effort Eliminated	Production Uptime	Enterprise AI Delivery

PROFESSIONAL SUMMARY

AI Engineer and Generative AI Lead with 3.5+ years architecting production-grade LLM applications, Agentic RAG systems, and multi-agent orchestration pipelines that generate quantifiable business outcomes. Built the full AI engineering stack — from advanced prompt engineering (few-shot, chain-of-thought, constitutional AI) and LLM fine-tuning (LoRA/PEFT on Llama-3 & GPT-4) to scalable cloud deployment on GCP Vertex AI, AWS (SageMaker, Bedrock), and Azure. Delivered \$500K+ in contract recoveries, 90% reduction in manual processing, and 99.9% production uptime across enterprise deployments. Expert in Hybrid RAG (RRF fusion, RHFL re-ranking), Knowledge Graphs, HITL validation, AI Governance, and FastAPI-based model serving. Led 7-person cross-functional teams across 3+ enterprise clients.

CORE TECHNICAL SKILLS

Generative AI & LLMs	GPT-4 / GPT-4o · Azure OpenAI · Llama-3 · Hugging Face Transformers · LLM Fine-tuning (LoRA, PEFT, QLoRA) · Constitutional AI · Function Calling · Embedding Models · Anthropic Claude
Advanced Prompt Eng.	Few-Shot · Chain-of-Thought (CoT) · Tree-of-Thought (ToT) · Self-Consistency · ReAct · Prompt Chaining · Structured Output Prompting · System Prompt Hardening · Jailbreak Mitigation
Advanced RAG	Hybrid Search (Dense + Sparse) · RRF (Reciprocal Rank Fusion) · RHFL Re-ranking · GraphRAG · VectorRAG · Knowledge Graphs (Neo4j, FalkorDB) · FAISS · Chroma · Milvus · Pinecone · Contextual Compression · HyDE
Agentic Orchestration	LangGraph · LangChain · CrewAI · AutoGen · Multi-Agent Systems · ReAct Planning · HITL (Human-in-the-Loop) Validation · Mem0 · Zep · Redis Checkpointers · Tool Use · MCP
Cloud & Model Serving	GCP Vertex AI · AWS SageMaker · AWS Bedrock · Azure OpenAI · FastAPI · Docker · Kubernetes · CI/CD · GitHub Actions · Streamlit · Model Serving · A/B Testing
LLMOps & Governance	Performance Monitoring Dashboards · Drift Detection · Prompt Versioning · Evaluation Frameworks (RAGAS, TruLens) · AI Governance · Responsible AI · PII Redaction · Audit Logging · Model Cards
Data & ML Engineering	Apache Spark · PySpark · Airflow · Snowflake · PostgreSQL · MongoDB · BERT · CNNs · XGBoost · TensorFlow · PyTorch · NER · Sentiment Analysis · Topic Modeling · SVM · Decision Forests
Languages & Frameworks	Python · SQL · PySpark · FastAPI · Streamlit · REST APIs · JSON Schema

PROFESSIONAL EXPERIENCE

AI Engineer & Generative AI Lead · Vcreatek

Jan 2023 – Present · Pune, MH

Technical Lead · 7 AI Engineers & Data Scientists · 3+ Enterprise Clients

► EPRMate — Production Agentic Compliance AI

- **Eliminated 90% of manual remediation effort** by architecting a multi-agent orchestration system on LangGraph + Azure OpenAI with ReAct reasoning loops, achieving 95% data completeness across enterprise-scale packaging compliance datasets.
- **Engineered HITL (Human-in-the-Loop) validation gates** at critical decision nodes, reducing erroneous auto-remediations by ~40% and satisfying enterprise audit requirements for regulated supply chain data.
- **Built fault-tolerant agent state management with Redis checkpointers and exponential backoff** — system maintained zero downtime across 6+ months of continuous production operation.

► Claim Recovery Engine — GenAI Document Intelligence

- **Recovered \$500K+ in supplier contract deductions** by replacing 100% manual review with an end-to-end LLM pipeline combining NER, LLM-based clause parsing, and structured output prompting on multi-format legal documents.

- ▶ **Reduced document reconciliation cycle from 5 days to under 2 hours** using FastAPI-served inference endpoints with async processing — pipeline operates with zero human touchpoints post-deployment.
- ▶ **Implemented AI Governance controls** including PII redaction, audit logging, and confidence-score thresholding to meet enterprise compliance requirements before production release.

► TEXTSPECTRUM — Zero-Code NLP Platform

- ▶ **Reduced time-to-insight by 60%** by enabling non-technical business users to self-deploy sentiment analysis and topic modeling pipelines via a no-code UI — eliminating ML engineering bottlenecks and 3-week delivery cycles.
- ▶ **Unified BERT + CNNs + LLMs in a hybrid architecture** for multi-domain text corpora, outperforming single-model baselines on cross-domain classification benchmarks by 12% F1.
- ▶ **Advanced prompt engineering (chain-of-thought, few-shot, self-consistency)** applied to LLM-layer classification reduced label hallucination by 22% vs. zero-shot prompting in production.

► Scalable Deployment, LLMops & AI Governance

- ▶ **Maintained 99.9% production uptime** across all AI applications deployed on GCP Vertex AI and Azure via FastAPI microservices, Docker/Kubernetes, and real-time performance monitoring dashboards with automated drift detection.
- ▶ **Improved domain-specific task accuracy by 15%** by fine-tuning Llama-3 and GPT-4 with LoRA/PEFT/QLoRA on specialized industrial datasets, with RAGAS-based evaluation pipelines for continuous quality assurance.
- ▶ **Established LLMops governance framework** covering prompt versioning, model cards, responsible AI policies, and drift alerting — adopted as the standard AI delivery template across 3 enterprise clients.
- ▶ **Secured multi-quarter C-suite buy-in at 3+ enterprises** by translating technical RAG and agentic AI architectures into ROI-anchored investment narratives, resulting in expanded AI program scope across all three accounts.

Data Science Intern · Vcreatek

Jun 2022 – Aug 2022 · Pune, MH

- ▶ **Processed 100K+ product reviews** using NLP (sentiment analysis, topic clustering) to build a GenAI-powered analytics dashboard — enabling data-driven product decisions for business stakeholders with no prior analytics tooling.

INDEPENDENT PROJECT

RAGentX — LangChain · LangGraph · OpenAI · Hybrid RAG · RRF · RHFL · FastAPI · Streamlit

Jan 2024 – Present

Ongoing open-source research project exploring production Agentic Hybrid RAG patterns

- ▶ **Built production Hybrid RAG system with RRF (Reciprocal Rank Fusion) and RHFL re-ranking** combining dense (FAISS) + sparse (BM25) retrieval with cross-encoder re-ranking — measurably reducing hallucination rates on open-domain research queries vs. naive dense retrieval.
- ▶ **Shipped full-stack agentic UI (FastAPI + Streamlit) with autonomous multi-tool API orchestration**, real-time answer streaming, and long-term memory (Mem0) — achieving 25% increase in simulated user engagement in A/B testing.
- ▶ **Implemented HyDE (Hypothetical Document Embeddings) and contextual compression** to improve retrieval precision by 18% on ambiguous queries, documented via RAGAS evaluation suite.
- ▶ **Architected agent state persistence with LangGraph checkpointers + Redis** enabling multi-session conversation continuity and graceful recovery from tool-call failures.

KEY ACHIEVEMENTS AT A GLANCE

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| <ul style="list-style-type: none"> ▶ \$500K+ contract deductions identified via LLM pipeline ▶ 90% reduction in manual compliance effort (EPRMate) ▶ 99.9% production uptime across all AI deployments ▶ 60% faster time-to-insight on NLP platform | <ul style="list-style-type: none"> ▶ 15% accuracy gain via LoRA/PEFT fine-tuning (Llama-3, GPT-4) ▶ 22% hallucination reduction with CoT prompt engineering ▶ 18% retrieval precision gain via HyDE + contextual compression ▶ AI Governance framework adopted across 3 enterprise clients |
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EDUCATION

PGDM — Research & Technical Business Analytics

2021 – 2023

Prestige Institute of Global Management